claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a4730fb5964c0c901.jsonl`
- SHA-256: `fa3080979cdd7967535c7d7d8b91f3aecd5ca85245992f024626cf0d4dd2b5c2`
- Source modified: `2026-06-16T03:28:24+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:27:08.869Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"The system detects hit moments primarily by finding direction changes in the shuttle trajectory (TrackNet's Peak Identification and Direction Identification Methods), then reconciles these with a separate swing-action detector through the Shot Refinement Algorithm to produce precise contact points. This is the exact mechanism missing from a pipeline that conflates 'shuttle moving' with 'rally in progress' ? trajectory direction-reversal/peak detection is the trajectory feature that marks a shot/contact event."

Source: https://www.mdpi.com/1424-8220/24/13/4372 (primary)
Supporting quote: "TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. ... The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:27:12.062Z)

I'll verify this claim by examining the source and checking whether the quote actually supports the specific mechanism described.

3. assistant (2026-06-16T03:27:14.807Z)

Let me fetch the primary source to verify the specific claims about the mechanism.

4. user (2026-06-16T03:27:17.872Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:27:19.660Z)

MDPI is blocking the fetch. Let me search for the paper and its details.

6. user (2026-06-16T03:27:27.572Z)

Web search results for query: "MDPI Sensors 2024 24 13 4372 badminton hit moment detection Shot Refinement Algorithm TrackNet Direction Identification"

Links: [{"title":"(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"},{"title":"MATCHMOTION: AN AI-POWERED MOBILE APPLICATION FOR AUTOMATED BADMINTON VIDEO ANALYSIS USING YOLOV8 OBJECT DETECTION","url":"https://www.researchgate.net/publication/404597971_MATCHMOTION_AN_AI-POWERED_MOBILE_APPLICATION_FOR_AUTOMATED_BADMINTON_VIDEO_ANALYSIS_USING_YOLOV8_OBJECT_DETECTION"},{"title":"AI-Driven Badminton Analysis: Enhancing Shot Detection and Player Performance Strategies | Technology","url":"https://www.devdiscourse.com/article/technology/3009448-ai-driven-badminton-analysis-enhancing-shot-detection-and-player-performance-strategies"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"}]

I found the research article you're looking for! This is a study published in MDPI Sensors in 2024.

Study Overview

The paper titled "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera" was published on July 5, 2024, by authors Yi-Hua Hsu, Chih-Chang Yu, and Hsu-Yung Cheng, with DOI 10.3390/s24134372.

Methodology

The study uses TrackNet, a deep neural network designed for tracking small objects, to extract the flight trajectory of the shuttlecock, and the YOLOv7 model to identify whether the player is swinging. A shot refinement algorithm is proposed to obtain the correct hitting moment by addressing detection misses and false detections from both models.

Results

The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches with 1582 rallies of Badminton World Federation match videos. This is a significant improvement compared to TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score.

The full article is available on the MDPI Sensors website at https://www.mdpi.com/1424-8220/24/13/4372

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-16T03:27:27.985Z)

Web search results for query: "badminton "hit moment" detection "Shot Refinement Algorithm" SRA shuttlecock trajectory swing action"

Links: [{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://doi.org/10.3390/s24134372"},{"title":"(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"},{"title":"Sensors (Jul 2024)","url":"https://doaj.org/article/492ce239283c4d5a9618446d651c03d5"}]

Based on the search results, I found relevant research on badminton hit moment detection and the Shot Refinement Algorithm:

Overview

The research proposes a novel method to extract shots in badminton games from a monocular camera. The approach consists of three main components:

Technical Approach

TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock. The YOLOv7 model is used to identify whether the player is swinging.

Shot Refinement Algorithm (SRA)

Since both TrackNet and YOLOv7 may have detection misses and false detections, the study proposes a shot refinement algorithm to obtain the correct hitting moment. The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment.

Performance Results

The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of the Badminton World Federation (BWF) match videos. This is a significant improvement compared to the use of TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score.

Shot Classification

Shots are categorized into seven primary types: clear, drive, smash, net, drop, lift, and push.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:27:31.391Z)

The search confirms the paper title and high-level structure. Now I need to verify the specific mechanistic claims, especially whether TrackNet's "Peak Identification Method" and "Direction Identification Method" exist as described, and what exactly the SRA combines. Let me fetch the open-access PMC version which should allow access.

9. user (2026-06-16T03:27:45.488Z)

Hit-Detection Mechanism Analysis

(1) Peak and Direction Identification Methods

The paper explicitly describes both methods:

Peak Identification Method: "The Peak Identification Method converts the shuttlecock position from (x, y) image coordinates to (fi, y), where fi represents the frame number."

Direction Identification Method: "To address this problem, TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock."

(2) SRA's Two Combined Modules

Yes, confirmed. SRA combines:
- **Shuttlecock trajectory module** using TrackNet's detection results (denoted HD-T)
- **Swing-action detector** using YOLOv7 (denoted HD-A for "Hit Detection by Action")

(3) Direction Method Quote

"TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. If there is a significant change in the direction of the shuttlecock, it is considered a hit."

(4) SRA's Objective Quote

"The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment."

(5) Trajectory Feature Confirmation

Confirmed. Hit moments are marked using trajectory direction reversals and peaks?the paper states hitting "primarily occur at the peak of this wave pattern" when converted to frame-space coordinates.

10. assistant (2026-06-16T03:27:49.094Z)

The PMC fetch confirms all the specific mechanistic claims. Let me do one more verification to ensure the claim's framing ? particularly whether "Peak Identification" and "Direction Identification" are attributed to TrackNet, and whether the trajectory direction-reversal is genuinely the contact-event feature ? is accurate and not an overreach. Let me check the arXiv hit-frame detection paper to cross-reference the trajectory-peak-as-contact concept.

11. user (2026-06-16T03:27:56.697Z)

Analysis of Hit Detection Methods in the Paper

Peak Identification Rationale

The paper explains the peak method as follows:

> "the hit moments primarily occur at the peak of this wave pattern"

The authors note that when shuttlecock position is converted from (x, y) to (frame number, y) coordinates, trajectories form wave-like patterns, with hits typically occurring at peaks where the y-coordinate reaches local maxima.

Direction Identification Method

Both methods are attributed to the original TrackNet work:

> "TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock"

The paper credits this approach to the prior TrackNet study, not presenting it as their own innovation.

Motivation for Adding Swing Detection

The authors explicitly identify trajectory-based limitations:

> "when the shuttlecock is occluded by the player's body or exits the field of view for a significant duration, it could increase the detection errors of the shuttlecock's trajectory"

They further note that relying solely on trajectories can lead to "substantial inaccuracies," motivating their proposal to combine trajectory analysis with YOLOv7-based action detection to resolve detection failures and false positives.

12. assistant (2026-06-16T03:28:08.674Z)

I now have enough verified evidence to assess the claim. Let me evaluate against the checklist.

Checklist assessment:

1. **Supported by quote?** Yes. The PMC open-access version confirms verbatim: "TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. If there is a significant change in the direction of the shuttlecock, it is considered a hit." It also confirms the Peak Identification Method (frame-vs-y wave pattern, hits at peaks), the SRA combining two modules (HD-T trajectory from TrackNet + HD-A swing action from YOLOv7), and the verbatim SRA objective sentence. The claim accurately describes the mechanism.

2. **Contradicting evidence?** None found. Multiple independent mirrors (PMC, ResearchGate, DOAJ) corroborate the same architecture and numbers (TrackNet-alone 72.3% F1 ? fused 90.5% F1).

3. **Source quality vs claim strength?** Peer-reviewed paper (MDPI Sensors 2024), open-access on PMC (NIH). The claim is a descriptive characterization of the paper's method, not an extraordinary empirical claim. Source quality is sufficient.

4. **Outdated?** 2024 paper, well within the 2022?2025 window. Current.

5. **Marketing/cherry-picked?** No ? it's a methods description from a peer-reviewed paper, not a press release.

One nuance to flag (a partial qualifier, not a refutation): The claim's editorializing ? "This is the exact mechanism missing from a pipeline that conflates 'shuttle moving' with 'rally in progress'" ? is the *voter's/researcher's* applicability framing, not something the paper asserts. The paper detects per-shot *hit/contact moments* (for shot classification), not rally start/end boundaries. Direction-reversal/peak as a contact-event trajectory feature is genuinely transferable to boundary work, but the paper itself addresses hit-frame detection, not the over-merge/rally-segmentation problem. The core factual claims about the mechanism are fully supported; only the relevance gloss is an interpretive extension. This does not meet the refutation bar (the substantive, sourced claims are all verified verbatim).

13. user (2026-06-16T03:28:18.343Z)

Structured output provided successfully

14. assistant (2026-06-16T03:28:24.521Z)

Verification complete. The claim is **not refuted** ? confidence high.

All substantive, sourced assertions are verbatim-confirmed against the open-access PMC mirror of the primary source (MDPI Sensors 2024, 24(13):4372): TrackNet's Direction Identification Method (sudden direction change ? hit) and Peak Identification Method (hits at peaks of the frame-vs-y wave), the SRA fusing a TrackNet trajectory module (HD-T) with a separate YOLOv7 swing-action detector (HD-A), and the exact SRA-objective sentence. Numbers (72.3% ? 90.5% F1) corroborate across three independent mirrors. The only soft spot is the voter's relevance gloss ("the exact mechanism missing from a pipeline that conflates 'shuttle moving' with 'rally in progress'"), which is interpretive ? the paper does hit/contact detection, not rally boundary segmentation ? but that is an extension, not a misrepresentation of the cited mechanism, so it does not meet the refutation bar.